BITS Pilani

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CE F231
Fluid Mechanics

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Lecture 6: Part 1
Pressure and its Measurement

Recap: Pressure Measurement Using Manometers

- Simple manometers → glass tube with one end connected to pressure measuring point and other end open to atmosphere
  - Piezometers
  - U-tube manometers
  - Single column manometers
    - Vertical single column
    - Inclined single column

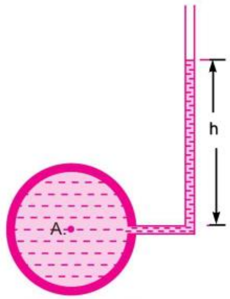
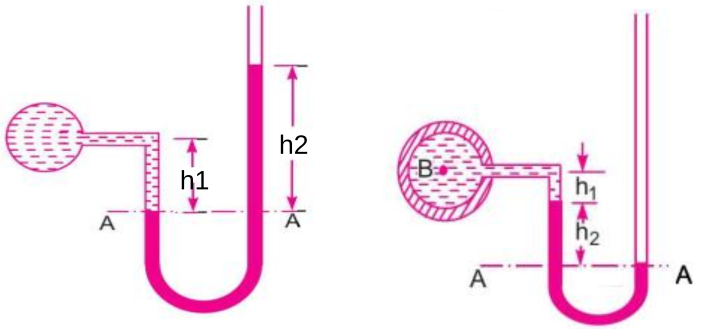


Fig. 2.8 *Piezometer.*



(a) For gauge pressure (b) For vacuum pressure

Fig. 2.9 *U-tube Manometer.*

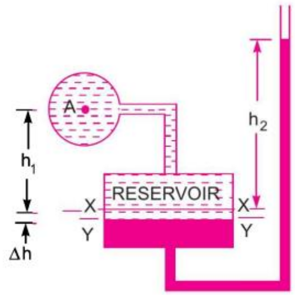


Fig. 2.15 *Vertical single column manometer.*

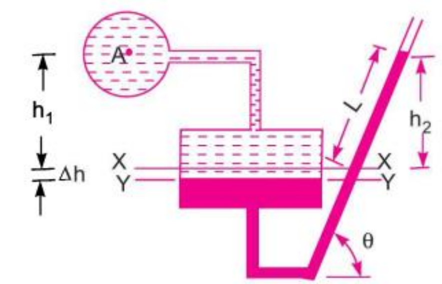
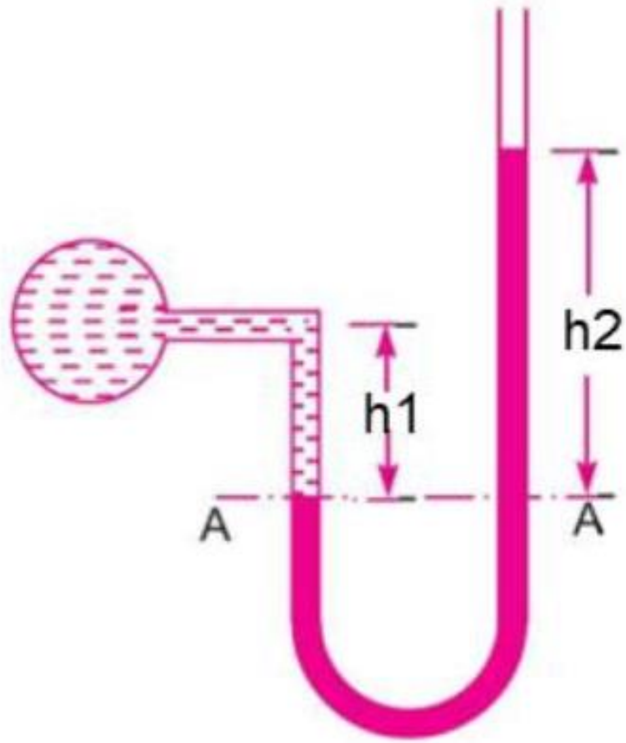


Fig. 2.16 *Inclined single column manometer.*

Bansal, R.K., 2005. *A textbook of fluid mechanics and hydraulic machines:(in SI units)*. LAXMI Publications, Ltd..

Recap: Pressure Measurement Using Manometers

Calculate the pressure in the pipe if h1 = 10 cm and h2 = 30 cm. Working fluid is water and manometric fluid is mercury



Bansal, R.K., 2005. *A textbook of fluid mechanics and hydraulic machines:(in SI units)*. LAXMI Publications, Ltd..

Recap: Pressure Measurement Using Manometers

- What is the initial level of MF?
- What is the final level of MF and WF interface after application of pressure?
- Why Δh is negligible after application of pressure?
- Pressure should be equated at which locations to be technically correct?
- At which locations pressures are equated accepting a small error in estimated pressure?

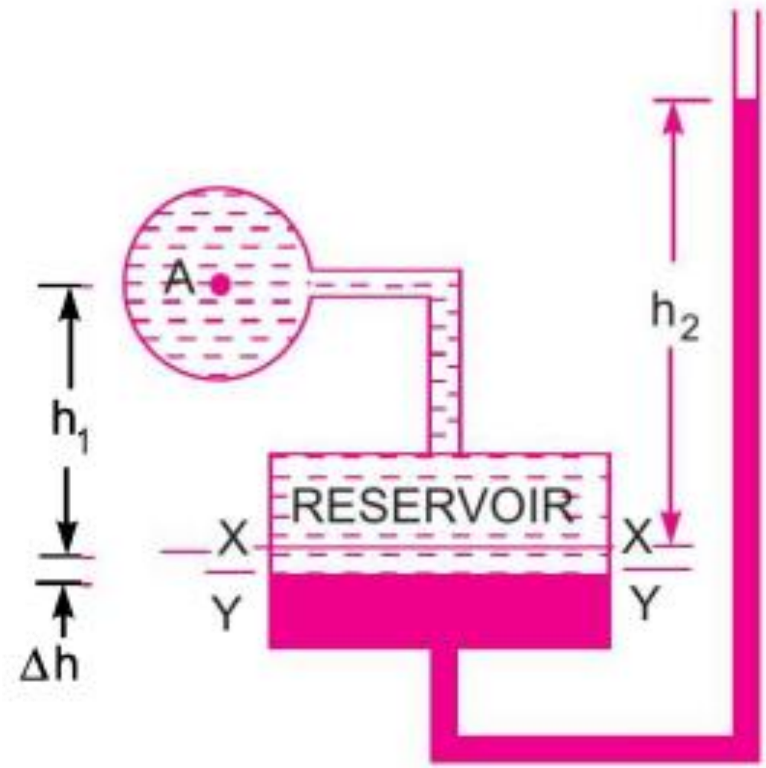
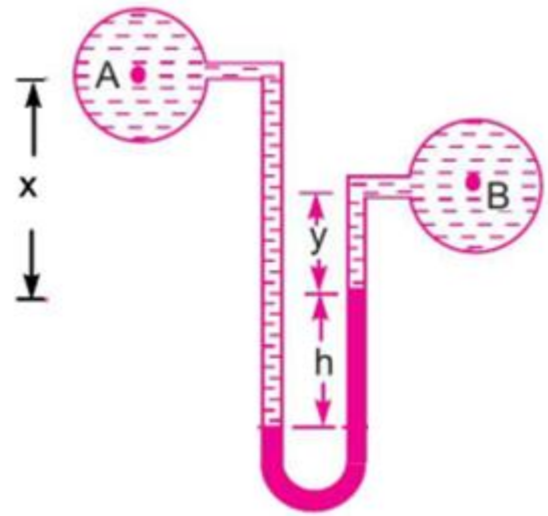


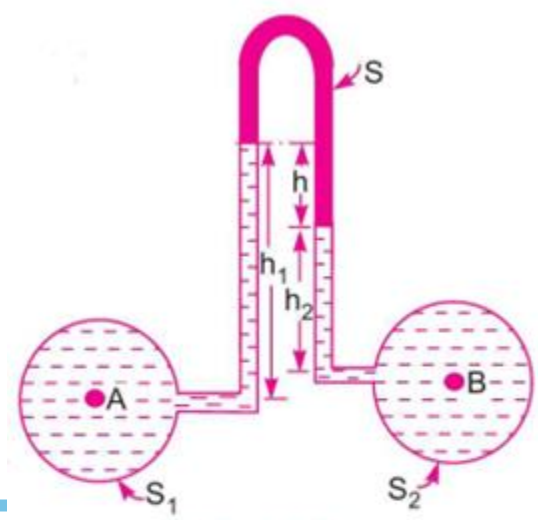
Fig. 2.15 *Vertical single column manometer.*

Bansal, R.K., 2005. *A textbook of fluid mechanics and hydraulic machines:(in SI units)*. LAXMI Publications, Ltd..

Pressure Measurement Using Manometers

- Differential manometers
  - U-tube differential manometer
    - Sp. gr. of manometric fluid > Sp. gr. of fluids whose pressure difference needs to measured
    - High pressure difference
  - Inverted U-tube differential manometer
    - Sp. gr. of manometric fluid < Sp. gr. of fluids whose pressure difference needs to measured
    - Low pressure difference

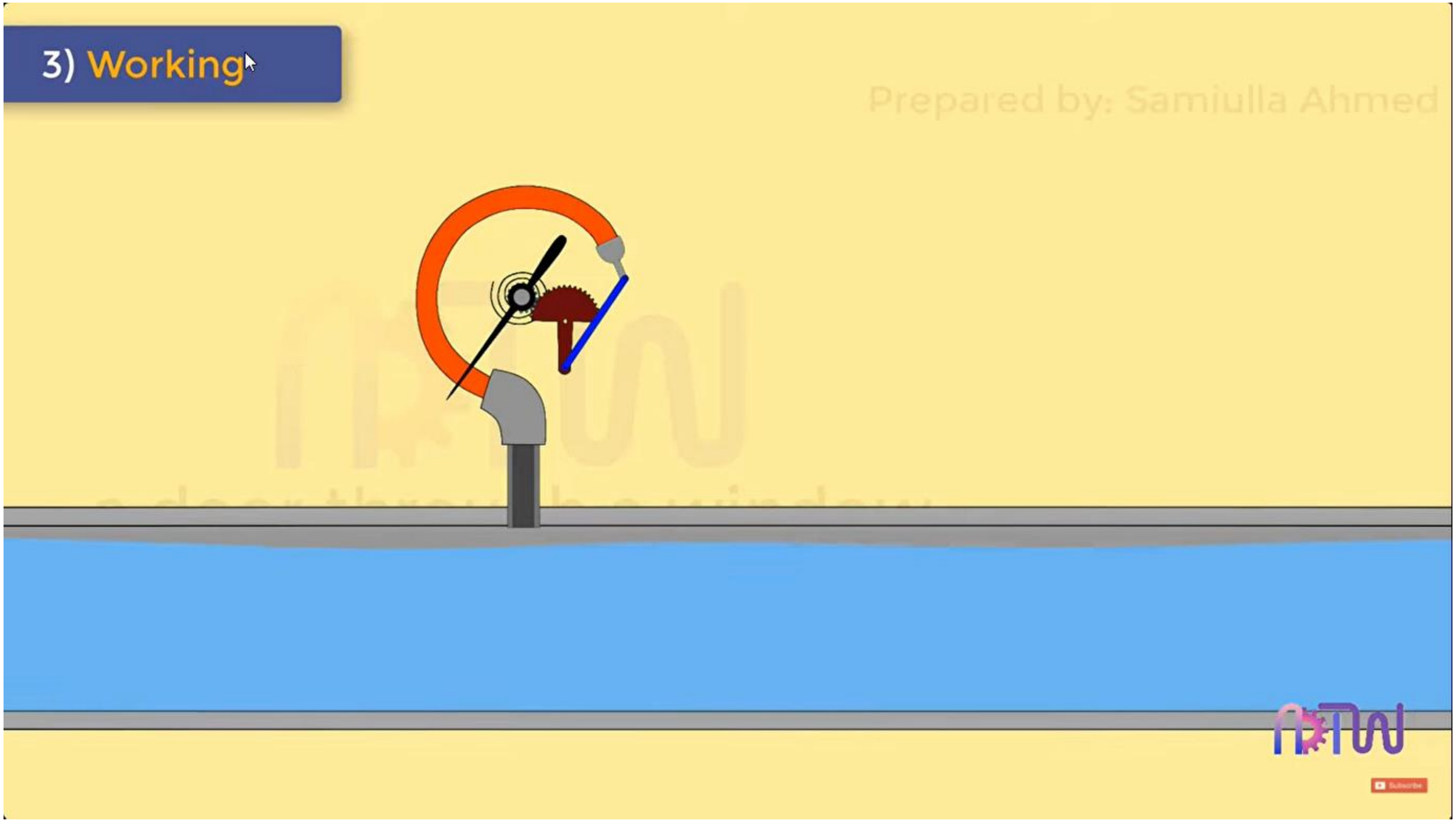




Bansal, R.K., 2005. *A textbook of fluid mechanics and hydraulic machines:(in SI units)*. LAXMI Publications, Ltd..

Mechanical Gauges

Bourdon tube pressure gauge (measuring spans 0-0.6 bar to 0-6000 bar)



Mechanical Gauges

Bourdon tube pressure gauge (measuring spans 0-0.6 bar to 0-6000 bar)

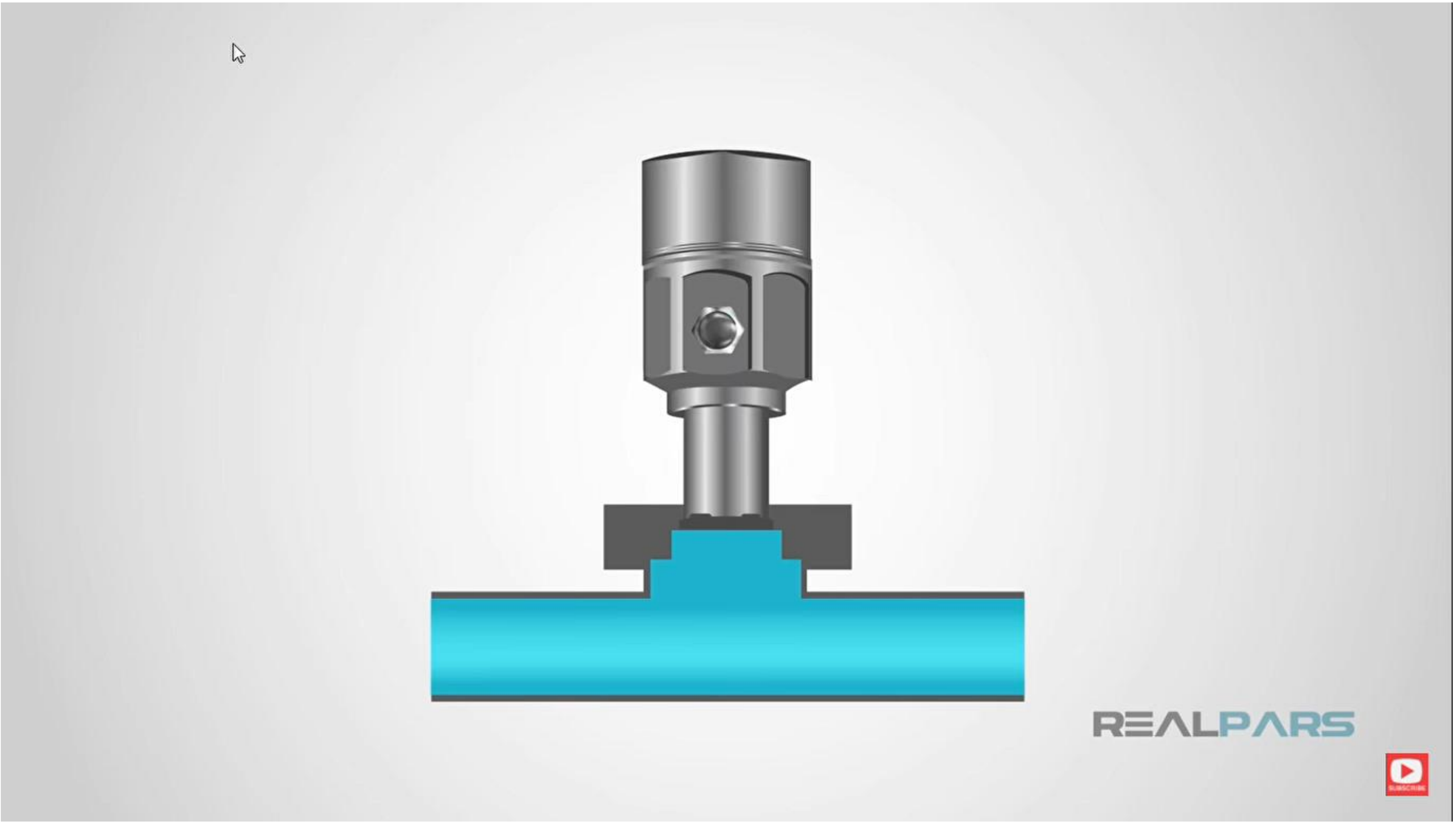
https://www.youtube.com/watch?v=9avIVZP1Ka4

Mechanical Gauges

Diaphragm pressure gauge (measuring low pressures 0-0.016 bar)

https://www.youtube.com/watch?v=JnPSeEW-EeA&t=90s

Pressure Sensors

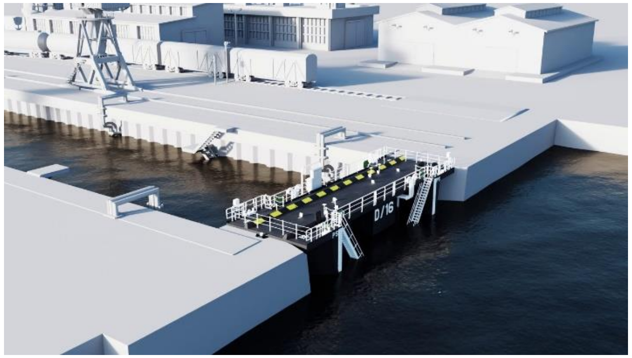


https://www.youtube.com/watch?v=iru8tRwS7Yc&t=135s

Lecture 6: Part 2
Hydrostatic Forces on Surfaces

Learning Outcome

- Learn how to estimate hydrostatic forces on surfaces
  - Forces on plane surfaces
    - Vertical
    - Horizontal
    - Inclined
  - Forces on curved surfaces



https://www.youtube.com/watch?v=VyJhWPduW5g

https://www.thehansindia.com/posts/index/Hans-Classroom/2017-06-09/Cofferdam/305433

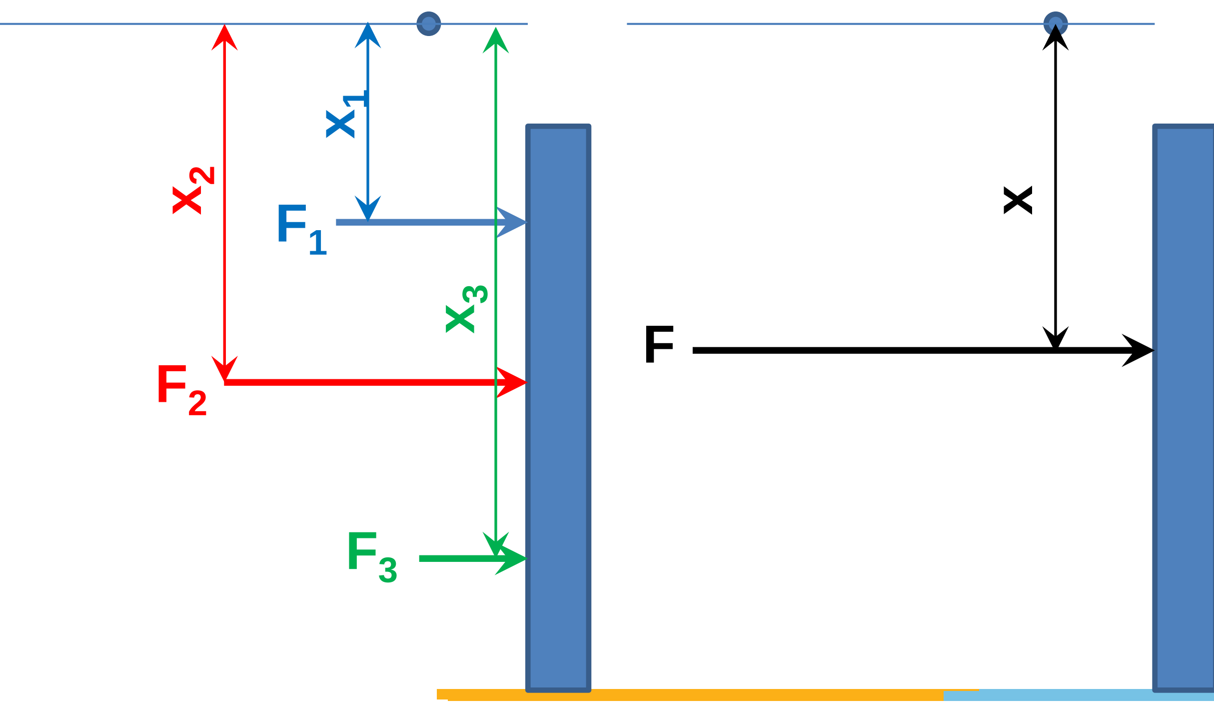
https://www.vortexhydradams.com/hydraulic-works/radial-gates/

Hydrostatic Forces on Surfaces

- Hydrostatic forces on surfaces
  - Estimate hydrostatic pressure forces
  - Determine centre of pressure
  - Magnitude, direction, and location of forces will be later used for designing water retaining structures
    - Dams
    - Dock gates
    - Lock gates
    - Sheet pile cofferdams
    - Other water retaining structures

Prerequisites

- The Principle of Moments, aka Varignon’s Theorem
  - The **Principle of Moments**, also known as **Varignon's Theorem**, states that the moment of any force is equal to the algebraic sum of the moments of the components of that force.



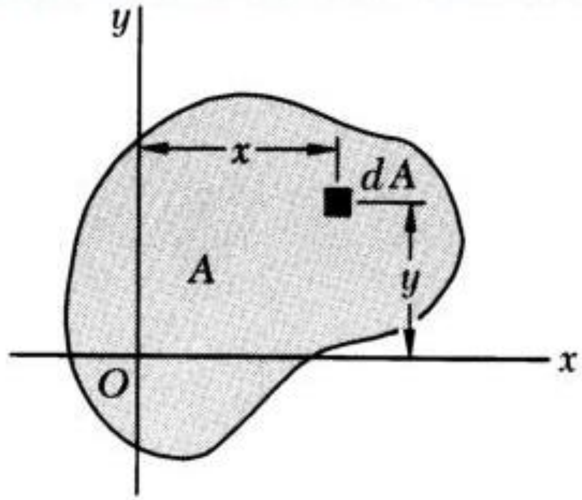
$$F = F_1 + F_2 + F_3$$

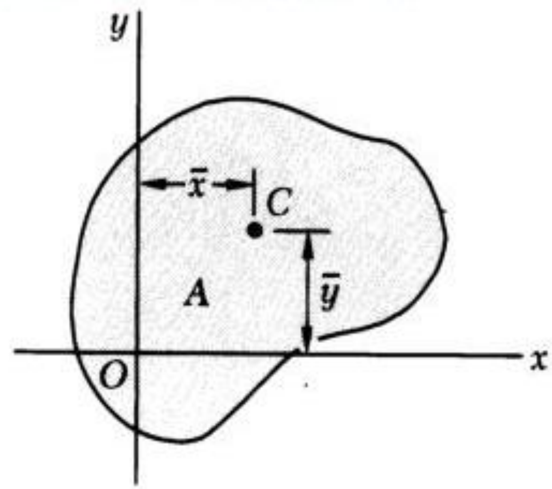
$$\boxed{Fx = F_1x_1 + F_2x_2 + F_3x_3}$$

Prerequisites

- First moment of area and centroid

A.1 First Moment of An Area; Centroid





First Moments of the Area *A* About the *x*- and *y*-Axis are Defined As

$$Q_x = \int_A y\,dA$$

$$Q_y = \int_A x\,dA$$

The *centroid of the area* A is defined as the point *C* of coordinates $\bar{x}$ and $\bar{y}$ which satisfy the relations

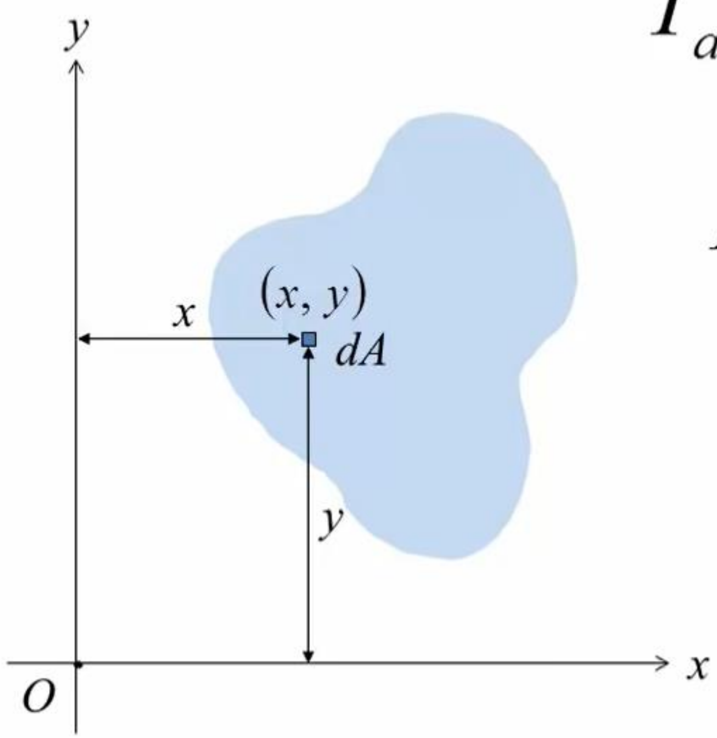
$$\bar{x} = Q_y / A$$

$$\bar{y} = Q_x / A$$

https://slideplayer.com/slide/2825728/

Prerequisites

- Second moment of area and area moment of inertia



$$I_{aa} = \int_A r_{aa}^{\,2}\, dA$$

$$I_x = \int_A y^2\, dA$$

$$I_y = \int_A x^2\, dA$$

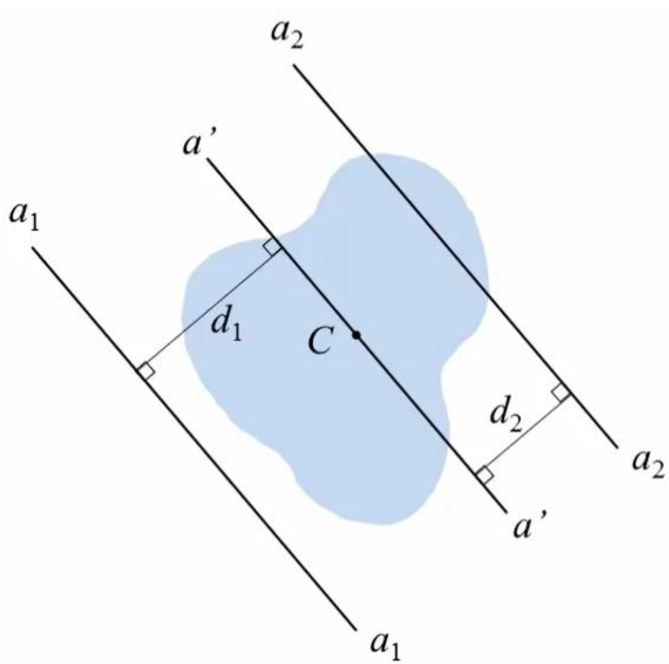
$$J_O = I_x + I_y$$

Polar moment of inertia

Prerequisites

➢ Parallel axis theorem

- The moment of inertia of an area with respect to any given axis is equal to the moment of inertia with respect to the centroidal axis plus the product of the area and the square of the distance between the 2 axes.



$\bar{I}_{a'}$ = moment of inertia about centroidal axis

$$I_{a_1} = \bar{I}_{a'} + Ad_1^2$$

$$I_{a_2} = \bar{I}_{a'} + Ad_2^2$$

Prerequisites

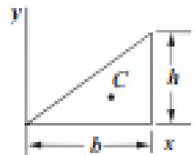
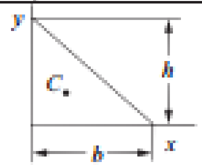
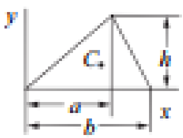
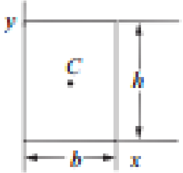
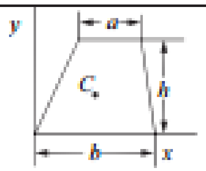
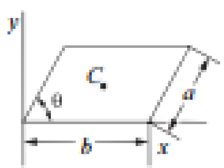
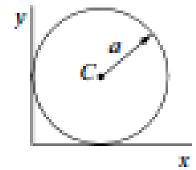
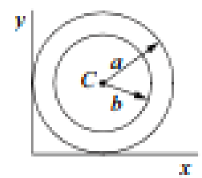
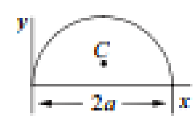
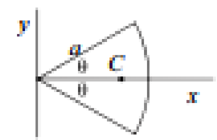
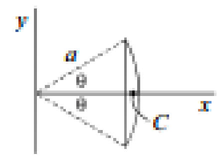
Figure	Area & Centroid	Area Moment of Inertia
	$A = bh/2$ $x_c = 2b/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = b^3h/36$ $I_x = bh^3/12$ $I_y = b^3h/4$
	$A = bh/2$ $x_c = b/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = b^3h/36$ $I_x = bh^3/12$ $I_y = b^3h/12$
	$A = bh/2$ $x_c = (a+b)/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = \left[bh\left(b^2 - ab + a^2\right)\right]/36$ $I_x = bh^3/12$ $I_y = \left[bh\left(b^2 + ab + a^2\right)\right]/12$
	$A = bh$ $x_c = b/2$ $y_c = h/2$	$I_{x_c} = bh^3/12$ $I_{y_c} = b^3h/12$ $I_x = bh^3/3$ $I_y = b^3h/3$ $J = \left[bh\left(b^2 + h^2\right)\right]/12$
	$A = h(a+b)/2$ $y_c = \frac{h(2a+b)}{3(a+b)}$	$I_{x_c} = \frac{h^3\left(a^2 + 4ab + b^2\right)}{36(a+b)}$ $I_x = \frac{h^3(3a+b)}{12}$
	$A = ab\sin\theta$ $x_c = (b + a\cos\theta)/2$ $y_c = (a\sin\theta)/2$	$I_{x_c} = \left(a^3b\sin^3\theta\right)/12$ $I_{y_c} = \left[ab\sin\theta\left(b^2 + a^2\cos^2\theta\right)\right]/12$ $I_x = \left(a^3b\sin^3\theta\right)/3$ $I_y = \left[ab\sin\theta(b + a\cos\theta)^2\right]/3 - \left(a^2b^2\sin\theta\cos\theta\right)/6$

Figure	Area & Centroid	Area Moment of Inertia
	$A = \pi a^2$ $x_c = a$ $y_c = a$	$I_{x_c} = I_{y_c} = \pi a^4/4$ $I_x = I_y = 5\pi a^4/4$ $J = \pi a^4/2$
	$A = \pi\left(a^2 - b^2\right)$ $x_c = a$ $y_c = a$	$I_{x_c} = I_{y_c} = \pi\left(a^4 - b^4\right)/4$ $I_x = I_y = \frac{5\pi a^4}{4} - \pi a^2b^2 - \frac{\pi b^4}{4}$ $J = \pi\left(a^4 - b^4\right)/2$
	$A = \pi a^2/2$ $x_c = a$ $y_c = 4a/(3\pi)$	$I_{x_c} = \frac{a^4\left(9\pi^2 - 64\right)}{72\pi}$ $I_{y_c} = \pi a^4/8$ $I_x = \pi a^4/8$ $I_y = 5\pi a^4/8$
 CIRCULAR SECTOR	$A = a^2\theta$ $x_c = \frac{2a}{3}\frac{\sin\theta}{\theta}$ $y_c = 0$	$I_x = a^4(\theta - \sin\theta\cos\theta)/4$ $I_y = a^4(\theta + \sin\theta\cos\theta)/4$
 CIRCULAR SEGMENT	$A = a^2\left[\theta - \frac{\sin 2\theta}{2}\right]$ $x_c = \frac{2a}{3}\frac{\sin^3\theta}{\theta - \sin\theta\cos\theta}$ $y_c = 0$	$I_x = \frac{Aa^2}{4}\left[1 - \frac{2\sin^3\theta\cos\theta}{3\theta - 3\sin\theta\cos\theta}\right]$ $I_y = \frac{Aa^2}{4}\left[1 + \frac{2\sin^3\theta\cos\theta}{\theta - \sin\theta\cos\theta}\right]$

Source: NCEES FE Reference Handbook

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